
EEMB 120 -- INTRODUCTION TO ECOLOGY

SUMMER 2020

LECTURE:	WEDS, THURS, FRI	online: Zoom 11:00-12:25
INSTRUCTORS:	ELIZABETH FORBES elizabethforbes@ucsb.edu	1117 NOBLE HALL BUT FOR NOW, ZOOM OFFICE HOURS: WEDS 12:30-1:30, OR BY APPT.
TA:	RUBY HARRIS-GAVIN rharrisgavin@ucsb.edu	ZOOM OFFICE HOURS: THURS 10-11 NOBLE 2003
REQUIRED READING:	PRIMARY LITERATURE POSTED ON GAUCHOSPACE	
OPTIONAL TEXTS:	1) ECOLOGY, 4TH EDITION WILLIAM D. BOWMAN, SALLY D. HACKER, MICHAEL L. CAIN 2) A PRIMER OF ECOLOGY (4 TH EDITION) 2008, N.J. GOTELLI 4) SERENDIPITY: AN ECOLOGIST'S QUEST TO UNDERSTAND NATURE (JAMES ESTES) – EXTRA CREDIT	
WEBSITE:	https://gauchospace.ucsb.edu/ (CONTAINS SECTION READINGS AND PODCASTS THAT YOU MUST DOWNLOAD)	
GRADING:	DISCUSSION	25 PTS. (15 PTS READING, 5 PTS ATTENDANCE, 5 PTS PRESENTATION)
	LECTURE	5 PTS ATTENDANCE
	MIDTERM	35 PTS.
	FINAL PROJECT	35 PTS.
	COURSE TOTAL	100 PTS.
	* 5 PTS EXTRA CREDIT AVAILABLE FOR BOOK REPORT; DETAILS BELOW	
EXAM DATES:	MIDTERM:	WEDNESDAY, JULY 13 TH (11:00-12:25)
	FINAL PROJECT:	FRIDAY, JULY 31 (IN CLASS DISCUSSION)
	OPTIONAL EXTRA CREDIT:	BY WEDNESDAY, JULY 29 TH (SUBMITTED ONLINE)
SECTIONS:	THURS.	1:00-2:20 PM
	THURS.	3:00 – 4:20 PM
	FRI.	9:00 – 10:20 AM
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Reading:

Chapters are in Bowman, unless otherwise marked. **Bold is required reading.** (Parenthetical material is optional). See week by week schedule below for details!

Attendance:

Attendance to lecture and section is mandatory, and attendance will be taken. For any lecture or section that you will not be able to attend, I request a reasonable explanation ahead of time. Missed lectures or sections without advance notice/explanation will result in a penalty to your grade.

If you have concerns related to this expectation to attend lecture and section (e.g. time zone or other time-sensitive restrictions) please do not hesitate to reach out to me (Elizabeth) to discuss them! Our goal is to optimize learning for the whole class, and in some cases accommodations may be necessary.

Participation:

We have a big class, but because it's online we have the opportunity to utilize creative means for participation. We will be requiring participation in both lecture and section, requirements which will involve combinations of discussion, small breakout groups, in-class polls, and out-of-class forums. Our goal is to encourage all kinds of learners to participate at different points.

How to access materials online:

All required (and some optional) materials will be posted online at our course Gauchospace. That's also where you'll find documents (like this one), and the link to our Zoom meetings. I will also be posting

Final project:

In lieu of a final exam this year, we will have a final project; details to come. In short, we'd love to see you apply some of the concepts from the course to a real-world example that interests you. Keep your eyes on Gauchospace, and we'll discuss it in class as the quarter gets going!

Code of conduct:

We will spend some time in the first lecture establishing, outlining, and agreeing upon a classroom code of conduct. I will ask you all to sign on to this code to help make our classroom an equitable, and welcoming place for all students to learn and challenge themselves in.

The elephant in the room: COVID-19

Let's get this out of the way: these are extraordinary circumstances to be living in, much less teaching and learning in. That's why it will be essential to establish and maintain an atmosphere of honesty and integrity in our classroom this summer. This course is fast, it is loaded with good stuff, and it's going to be challenging; however, we will start by checking in, and will continue to check in throughout our six weeks together, to make sure everybody's learning objectives are being met. Take this as your first of many invitations to speak to us (myself or Ruby) if you have any specific concerns. Our (metaphorical, of course) doors are open.

Campus resources:

Resources at your disposal this summer:

- CARES: 24-hr advocacy line (805) 893-4613 [Zoom meetings available],
- CAPS: new and initial appts scheduled at (805) 893-4411 [Zoom meetings available]
- Health and Wellness: <http://wellness.sa.ucsb.edu>
- Student Health Center: telehealth, in person appointments, mailing of prescriptions
<http://studenthealth.sa.ucsb.edu>

LECTURE:

WEEK OF: TOPIC:

READING ASSIGNMENT:

June 22	Introduction to Ecology & physical environment ▪ <i>What are you going to get out of this class?</i>	Ch. 1 (2), Diamond 1986
June 25	Physiological Ecology: ▪ <i>How do organisms survive in their environment?</i> ▪ <i>What kinds of limits does an environment put on species?</i>	Ch. 4 (2,5) Williams et al 2014
June 26	Behavioral Ecology ▪ <i>How do organisms adapt their behaviors in response to their environment?</i> ▪ <i>What ecological factors drive changes in behavior?</i>	Ch 8, Tinker et al 2008

June 29	Population Ecology I- Exponential	Ch 9,10, Finkelstein et al 2012 , Gotelli 1
July 2	Population Ecology I – Exponential (continued) ▪ <i>What do we mean when we refer to species population growth?</i> ▪ <i>What are the factors that determine if a population is growing or not at any point in time?</i>	
July 3	NO CLASS: Independence day holiday	

July 6	Population Ecology II – Age Structured III ▪ <i>How can we use the life stages and ‘survivorship schedule’ of a species to help us predict population growth?</i>	Ch 10, Fujiwara Caswell 2001 , Gotelli 3
July 9	Population Ecology III - - Logistic ▪ <i>What factors (species or environmental) might limit population growth?</i>	Ch 10-11, Bradshaw and Brook 2014 , Gotelli 2
July 10	Species Interactions: Competition ▪ <i>How might competition for resources between species influence population growth of those species?</i>	Ch. 14

July 13	** Exam 1 Weds July 17 11-12:25***	TBD format
July 16	Species Interactions: Predation ▪ <i>How does predation impact populations of both predators and prey?</i> ▪ <i>How does predation impact species behavior?</i>	Ch. 12, Borer et al 2005
July 17	Species Interactions: Parasitism ▪ <i>What is parasitism compared to other kinds of species relationships?</i>	Ch. 13, Kuris et al 2008 , Gotelli 6

- *How do parasites impact populations, behaviors, and species interactions?*

July 20	Species Interactions: Complex interactions ▪ <i>What kinds of interactions can species have?</i> ▪ <i>What kinds are feeding-based, and what kinds are non-feeding?</i>	Ch. 21 (16), Trussel et al 2017, Palmer 2008
July 23	Diversity 1 – Patterns ▪ <i>How are species distributed globally and why?</i>	Ch 18, Condamine et al 2012 , Gotelli 7,9
July 24	Diversity II - Consequences ▪ <i>How do we describe patterns of species distribution?</i>	Ch 19, Hooper et al 2012
July 27	Climate Change ▪ <i>How does a changing climate impact species, species interactions, populations, and distributions?</i>	Pinsky et al 2019
July 30	Conservation Ecology ▪ <i>What is conservation ecology?</i> ▪ <i>Based on what you know about species, interactions with each other and their environments, populations, and distributions, what are the strategies you'd favor for conservation, and why?</i>	Ch 23-24, Doak 2013
July 31	*** Final Project Due***	Final Project due, wrap-up survey in class